Using OpenCV for Computer Vision Techniques

OpenCV, also known as Open Source Computer Vision Library, is a cross-platform library of computer functions, developed by Intel with the aim to achieve real-time computer vision.

Computer vision is a domain that has many algorithms that are very time consuming and complex in terms of core programming. The core areas in computer vision include 2D and 3D features, facial recognition systems, gesture recognition, human–computer interaction (HCI), mobile robotics, motion understanding, object identification, segmentation recognition, stereopsis stereo vision (depth perception from two cameras), structure from motion (SFM), motion tracking, augmented reality, pattern recognition, scene reconstruction, decision making and many others.

Even though computer vision algorithms can be implemented or simulated in MATLAB, SciLab and VC++, yet their rapid implementation will be an issue until we have a specific tool for computer vision techniques. So let's look at a specific open source tool called OpenCV (Open Source Computer Vision Library) that is used to implement computer vision algorithms and techniques. OpenCV is widely used in research laboratories and in industry to achieve explicit and efficient results.

OpenCV *[http://opensource.org]* is an open source library of the functions primarily related to real-time computer vision techniques. According to SourceForge.net, OpenCV has more than 40,500 downloads per week. It has been developed by Intel and is currently supported by Willow Garage and Itseez. The software suite is free for use, is cross-platform and is under open source BSD licence. The software library focuses very efficiently on real-time image processing and computer vision algorithms.

The fame of OpenCV can be gauged from the fact that it has a huge community of users and an estimated 6 million downloads. Along with well-established companies like Google, Yahoo, Microsoft, Intel, IBM, Sony, Honda and Toyota that employ the library, there are many start-ups such as Applied Minds, VideoSurf and Zeitera, which make extensive use of OpenCV. It has been deployed for tasks that range from stitching streetview images together, detecting intrusions in surveillance videos in Israel, monitoring mine equipment in China, helping robots navigate and pick up objects at Willow Garage, and detecting swimming pool drowning accidents in Europe,